

First manual execution:

Lowest length path between s and t using BFS (forward) from S.

Node dictionary

key	value
0	$\{1, 2, 3\}$
1	$\{2, 4\}$
2	$\{0, 3, 4\}$
3	$\{4\}$
4	$\{1\}$

S=0, t=4	x	y	queue	visited	dist-dictionary	prev-dictionary
Initialization	-	-	<u>0</u>	$\{0\}$	$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 \end{bmatrix}$
Iteration 1.1	0	1	<u>1</u>	$\{0, 1\}$	$\begin{bmatrix} 0 & 1 & 1 & 1 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 1 & 1 & 1 \end{bmatrix}$
Iteration 1.2	0	2	<u>2</u>	$\{0, 1, 2\}$	$\begin{bmatrix} 0 & 1 & 1 & 1 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 1 & 1 \end{bmatrix}$
Iteration 1.3	0	3	<u>3</u>	$\{0, 1, 2, 3\}$	$\begin{bmatrix} 0 & 1 & 1 & 1 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 & 1 \end{bmatrix}$
Iteration 2.1	1	vis.	<u>2</u>	$\{0, 1, 2, 3\}$	-	-
It 2.2	1	4	<u>4</u>	$\{0, 1, 2, 3, 4\}$	$\begin{bmatrix} 1 & 1 & 1 & 1 & 2 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \end{bmatrix}$
		Stop				

The key path built from prev: $\begin{bmatrix} 0 & 1 & 2 & 3 & 4 \end{bmatrix}$ beginning with t=4.
 prev[4] = 1, prev[1] = 0, prev[0] = -.
 rev path: [4, 1, 0].
 path: [0, 1, 4].
 length = dist[4] = 2

Second manual execution:

S=3, t=0

S=3, t=0	x	y	queue	visited	dist dict.	prev dict
Initialization	-	-	<u>3</u>	$\{3\}$	$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 \end{bmatrix}$
It 1.1	3	4	<u>4</u>	$\{3, 4\}$	1	3
It 2.1	4	1	<u>1</u>	$\{1, 3, 4\}$	2	4
It 3.1	1	2	<u>2</u>	$\{1, 2, 3, 4\}$	3	1
It 3.2	1	vis.	<u>2</u>	$\{1, 2, 3, 4\}$	-	-
It 4.1	2	0	<u>0</u>	$\{0, 1, 2, 3, 4\}$	4	2
		Stop				

prev[0] = 2, prev[2] = 1, prev[1] = 4, prev[4] = 3
 1) rev path: [0, 2, 1, 4, 3]
 2) path: 3 → 4 → 1 → 2 → 0.
 Len = 4.

Minimum-length paths on the large graphs

graph1k

From terminal:

Lowest length path: 1 -> 5 -> 487 -> 175 -> 699 -> 624 -> 100

Length: 6

Lowest length path: 100 -> 416 -> 354 -> 865 -> 109 -> 1

Length: 5

So:

- 1 -> 100: 1 -> 5 -> 487 -> 175 -> 699 -> 624 -> 100, length 6
- 100 -> 1: 100 -> 416 -> 354 -> 865 -> 109 -> 1, length 5

graph10k

From terminal:

Lowest length path: 1 -> 3300 -> 2607 -> 523 -> 6311 -> 5359 -> 9794 -> 5173 -> 100

Length: 8

Lowest length path: 100 -> 2398 -> 3054 -> 5232 -> 8217 -> 2478 -> 7151 -> 1

Length: 7

So:

- 1 -> 100: 1 -> 3300 -> 2607 -> 523 -> 6311 -> 5359 -> 9794 -> 5173 -> 100, length 8
- 100 -> 1: 100 -> 2398 -> 3054 -> 5232 -> 8217 -> 2478 -> 7151 -> 1, length 7

graph100k

From terminal:

Lowest length path: 1 -> 17024 -> 27471 -> 14969 -> 3075 -> 4156 -> 32753 -> 14973 -> 100

Length: 8

Lowest length path: 100 -> 44340 -> 54527 -> 6606 -> 53263 -> 95930 -> 98655 -> 58288 -> 1

Length: 8

So:

- 1 -> 100: 1 -> 17024 -> 27471 -> 14969 -> 3075 -> 4156 -> 32753 -> 14973 -> 100, length 8

- 100 -> 1: 100 -> 44340 -> 54527 -> 6606 -> 53263 -> 95930 -> 98655 -> 58288 -> 1, length 8